

Affordable Intelligent Robot for Railway Track Surveillance

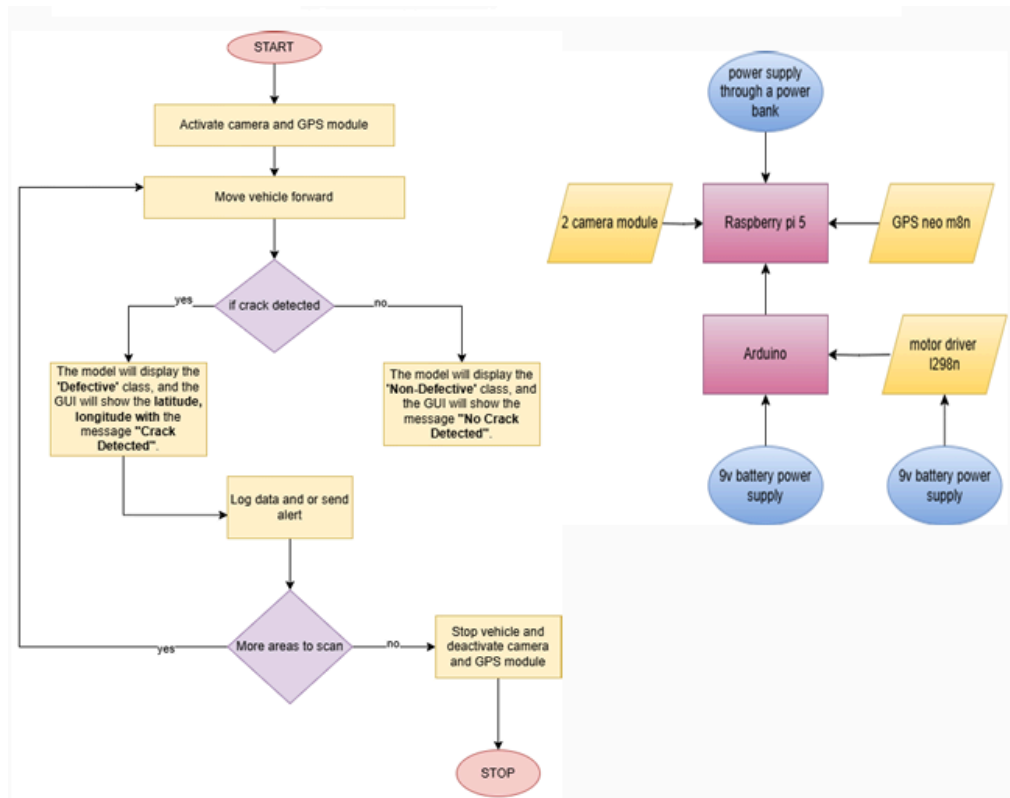
The patent is in the filing stage.

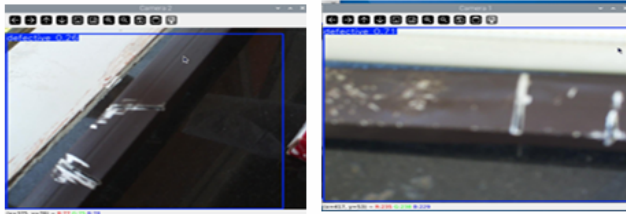
Railway track cracks pose serious safety risks and operational losses, while existing inspection methods (Inspection car and manual inspection) are expensive, slow, and error-prone. There is a need for an affordable, automated, and real-time inspection solution for large-scale deployment. The project is in the first phase.

This project focuses on crack detection using computer vision and deep learning. When a bot moves along the custom prototype railway track and uses **two Pi cameras** to monitor both rails. When a crack is detected, the system **matches the crack bounding box time with GPS latitude and longitude** and displays the exact crack location on the **GUI**.

I was primarily responsible for **custom dataset creation and model development**. Railway track images were collected under real-world conditions and augmented using Roboflow (flipping, blurring, and brightness variations), expanding the dataset to **9,000+ images**. The object detection model was trained using **YOLOv8n** with the **AdamW optimizer** and early stopping for improved convergence. Model performance was assessed using metrics and results were Precision: 0.988, Recall: 0.997, **mAP@0.5: 82.2**, and **mAP@0.5-0.9: 62**, demonstrating strong accuracy and suitability for real-time edge deployment.

Hardware Setup: Custom-built environment, mobile robot with motor driver, DC motors, GPS module, Arduino, and Raspberry Pi 5.





Camera feed showing defective region



Railway crack detection bot and environment prototype

?? Railway Crack Detection Monitor

?? Live Crack Detection Feed

Time	Status	Latitude	Longitude
2025-04-13T12:44:10.1	Crack Detected	18.5433	73.9122
2025-04-13T12:44:12.1	Crack Detected	18.5433	73.9122
2025-04-13T12:44:14.1	Crack Detected	18.5433	73.9122
2025-04-13T12:44:15.1	Crack Detected	18.5433	73.9122
2025-04-13T12:44:17.1	Crack Detected	18.5433	73.9122
2025-04-13T12:44:18.1	Crack Detected	18.5433	73.9122
2025-04-13T12:44:19.1	Crack Detected	18.5433	73.9122
2025-04-13T12:44:25.1	Crack Detected	18.5433	73.9122
2025-04-13T12:44:26.1	Crack Detected	18.5433	73.9122
2025-04-13T12:44:28.1	Crack Detected	18.5433	73.9122
2025-04-13T12:44:27.1	Crack Detected	18.5433	73.9122
2025-04-13T12:44:28.1	Crack Detected	18.5433	73.9122
2025-04-13T12:44:30.1	Crack Detected	18.5433	73.9122

?? Export CSV Crack Detected

This is a GUI displaying location of the crack detected

